

[Working title] Who is the Pronoun Pro?

Comparing LLMs and Humans at Number-Disambiguation of *they*

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1 Introduction

Large Language Models (LLMs) have not only permeated wider society, they have also begun changing the methodological landscape in the humanities and linguistics more specifically. The promise of the team behind GPT-3 was that Natural Language Processing (NLP) as a whole would shift “to using task-agnostic pre-training and task-agnostic architectures” Brown et al. (2020, 1), i.e. general-purpose language models able to perform a wide range of tasks without task-specific training. Indeed, LLMs have found a wide variety of uses: LLMs have been used for X, Y, and Z in linguistics where their performance was found to be so and so.

The present work was performed in the context of an older effort starting in my Master’s thesis (2022). The then-new GPT-3 was tested for its performance at disambiguation uses of *they* between plural and singular (see below). Coreference resolution as a field precedes LLMs, though it was always limited by the many challenges posed by the nature of coreference, such as real-world knowledge required for textual understanding and context-dependency. What is more, English pronouns have shifted in recent years, most notably with the extension of *they* as a personal pronoun for people of non-binary gender, but also with the growing visibility of neologistic pronouns. LLMs showed promise because they exhibited context-sensitive performance in several areas. While its performance with singular *they* in the dataset of the Master’s thesis was unsatisfactory, the same test was later repeated using newer models. When Anthropic’s LLM Claude Sonnet reached both recall and precision of over 80% on the same dataset in late 2024, it was hypothesized that LLMs had reached human-like performance in this task. To assess this reliably and accurately, the Research Question was formulated: “To what extent

are number-annotations of authentic uses of *they* by LLMs reliable when compared to trained human annotators?”

To answer this question, three levels of performance quality were defined, as expressed by the following three hypotheses:

- Hypothesis 1: The most accurate number-disambiguator of *they* is going to be an LLM.
- Hypothesis 2: The most accurate LLM at number-disambiguating *they* is going to perform at a higher accuracy than the mean human accuracy.
- Hypothesis 3: The performance of the most accurate LLM is not going to be significantly worse at rare and innovative uses of *they* than the mean human performance.

The three hypotheses were defined to capture the extent to which LLM performance could potentially be comparable to that of humans. Even if hypothesis 1 is not confirmed by the results, hypothesis 2 would still attest that LLMs largely outperform humans. If that were not confirmed either, hypothesis 3 would still justify the use of LLMs for semantically-complex tasks such as resolution of innovative pronouns. Additionally, the present work also allowed to assess more generally and qualitatively in what ways LLMs are human-like and in what ways they are not; and if, perhaps, any revealing patterns could be found in the recorded behaviors of LLMs and humans. This is expressed in the additional research question: “In what ways are LLMs human-like and in what ways are they not when it comes to number-disambiguation of *they*?”

2 Background

Point out that the term LLMs is not synonymous with decoder-only models but that the former will be used for simplicity and because the models tested in this work are all decoder-only. (ref to Arslan, Erol, and Eryigit 2026)

As this task is supposed to represent any task relevant in the field of linguistics in which the mere surface level is not enough for correct processing, the following background is not strictly relevant. It will help to contextualize the present research and justify the task design.

- Haq et al. (2025) also employ commercial and non-commercial LLMs. They don’t show that either is inherently better but that model size is an important factor. Commercial models tend to be larger as they are trained on more data, though open-weight models have caught up in that regard (# SOURCE).
- Arslan, Erol, and Eryigit 2026 investigate Coreference Resolution with LLMs and fine-tune models for it. => I do not even fine-tune and reach human-like results They investigate zero anaphors (i.e. omitted pronouns) and find that LLMs perform well on them, which is relevant for the present research as it shows that LLMs show “a deep understanding of the contextual focus”. Their work crosses linguistic boundaries which

poses interesting challenges. My work does not do that but instead contains innovative language which tends to be rare in more general-purpose datasets.

2.1 Coreference resolution

Describe task with reference to Jurafsky and relate its history with Gan et al. 2025.

As with all NLP tasks, LLMs have changed the landscape of coreference resolution. The field has shifted from requiring task-specific training material to being able to use general-purpose LLMs.

More recent history of CRR in Gan et al. 2025.

The present research is not about coreference resolution as a whole but about a specific aspect of it, i.e. the number-disambiguation of *they* which works through the identification of the referent.

Point out that in this case, the mention detection (ref to textbook) is precluded by our interest in the pronoun *they* and that the task is thus reduced to a classification task of number-disambiguation of *they*.

The task at hand takes a risk by skipping the identification of the referent, making its output less useful for downstream tasks. However, the task is still relevant for the field of linguistics as it allows to study the phenomenon of *they* in a more controlled way and to assess the extent to which LLMs can handle semantically complex tasks.

It has been shown repeatedly that LLMs encode gender bias (Bolukbasi et al. 2016, Lucy and Bamman 2021, Lee et al. 2025 section 5, Shore et al 2025 section ??) Gender bias could still represent a hurdle for LLMs in this task, as the number-disambiguation of *they* is often related to the gender of the referent.

2.2 Three kinds of *they*

(plural, generic, singular)

The pronoun *they* is generally used as a plural pronoun of the third person, meaning that it refers to groups of people or things that are not directly addressed. However, it is rather well-known that its usage is not limited to plural reference but that a “singular *they*” exists as well. The reason that this grammatical ambiguity is so well-known is that feminist discourses of the latter half of the 20th century drew attention to contexts in which writers would use the pronoun *he* despite the context not strictly mandating a masculine pronoun. This is known as “generic *he*”. To avoid linguistic and thus cognitive erasure of women, the use of the *they* in such contexts was advocated instead. This discourse seamlessly flowed into a new era of gender politics that drew attention to at least two new pronoun-related concerns: 1) the use of *they* to mask gender wherever it is not strictly necessary to reveal it, and 2) the use of *they*

as a personal pronoun for people of non-binary gender. The first concern is perhaps not as prominent as the second, and speakers are perhaps not as aware of it. The following examples illustrate how this tendency reveals itself:

#TODO EXAMPLES

Example N comes from a public figure in science communication of mathematics, Matt Parker. It is perhaps surprising that he avoids using the gendered pronoun *he* in reference to a known male mathematician. In Example M, Parker avoids using a gendered pronoun when talking about a person who the audience has not seen yet, and therefore does not know their gender. In my personal correspondence with Parker, he revealed that this was indeed a deliberate choice that he decided to make in his public speaking since he did not want “the audience to form any opinions based on that”. Parker continued to explain that examples such as N came about due to unconsciously learning to use *they* as a default. Such use of *they* corresponds to the self-reported use in Konnelly and Cowper. Example(s) O (etc) show that the masking of gender also happens in other, less monitored, contexts, indicating that a more general language change is underway, albeit most likely motivated by the first concern mentioned above. The second concern, the correct gendering of non-binary individuals, is more straightforward and is best explained by linguists and other writers who are more directly involved in the discourse around non-binary gender, such as X and Y.

What both concerns mean for the present research is that they make the use of truly singular, and not merely generic, *they* more likely. It can also, and perhaps more usefully, be called “referential singular *they*”. It was alluded to above that *they* was propagated to replace generic *he* in the second half of the 20th century, and indeed it has been found that *they* has since expanded to more contexts that had not been considered before (#REFS). To better grasp the changing use of *they*, synchronically and diachronically, it is thus necessary to highlight the distinction between these two innovations of *they* use.

Generic singular *they* is the use of *they* in contexts that are grammatically singular but generic in meaning. “Generic” means that (#TODO: dictionary-sourced definition). Grammatically speaking, generic statements can make use of singular or plural forms. The former can be called “instantiation” (#TODO, reference Newman here) and could take the form of expressions such as *the average worker* or *anyone in their right mind*. Using *they* to refer to such expressions was not always regarded as correct, but a rich history of such usage is well-documented (#REF). Judging generic singular *they* as (in)correct was always related to positions about linguistic gender-fairness, as the following examples exemplify:

In these examples, the phenomenon of instantiation is to some degree problematic as a male pronoun instantiates a male person in most people’s minds, potentially exacerbating an already overly male-centric worldview. And indeed, it is well attested that opposition or promotion of generic singular *they* is often motivated by ideological stances on gender-fairness (#REF). Most notably, the mid-19th century saw a wave of androcentrist grammatical prescriptivism that led to the decline of generic singular *they* in formal writing, and the late 20th century saw a wave of feminist advocacy that led to the comeback of generic singular *they* in formal

writing. Some scholarship suggests that the use generic singular *they* in informal speech and literary writing was generally unaffected by these trends.

Singular *they* can be conceived as... #TODO

Both generic and specific singular *they* can be differentiated further. For example, the subject of generic statements can carry semantic gender. For illustration, *the average worker ... they* is likely less controversial than *the average mother ... they*. Specific singular *they*, especially since it is an innovative form, could be found in an array of different contexts. One important distinction can be based on speaker knowledge: *my friend ... they* might be more likely to be used when the hearer does not know the referenced friend’s gender, or perhaps when the hearer will never learn that friend’s gender. Both cases might still be less likely than cases in which the speaker does not know the referenced person’s gender, such as *the backstage manager ... they*. And most saliently, current discourse around singular *they* is focused on the use of *they* as a personal pronoun for non-binary individuals, the use of which is likely to correlate with knowledge about and acceptance of gender diversity (Remsö et al 2026).

Thus, the phenomenon usually subsumed under the term “singular *they*” is actually a heterogeneous set of uses that differ in their discursive function and likely also in their patterns of use over time. This is not a new insight and others have attempted to differentiate all possible and/or plausible uses of *they* in the literature. One particularly thorough attempt is that of Newman (1994, 1997), who situates every use of *they* along several axes (#TODO) and thus really only lacks non-binary *they* as Newman was seemingly unaware of the possibility of such a use. (#TODO: check that one footnote in which he talks about transvestites etc)

However, for the present purpose a balance needs to be struck between the granularity of the typology and the feasibility of annotating it with LLMs. Gauging speaker/hearer knowledge and other nuances of each context is not relevant when specific singular *they* is still fairly rare on the whole and is likely even absent in many datasets. For most research intents, the task of annotating instances of *they* is really only about differentiating between plural uses and any of the innovative ones. The latter can then be studied closely by a human researcher. However, I maintain that some depth is necessary when analyzing in order to differentiate between generic and specific singular *they* for the simple reason of scalability. To provide a broad picture of how *they* has changed broadly over time across different contexts and speaker groups, close analysis of each instance is not necessary or feasible. Instead, it would be preferable to blindly trust the annotator(s). Based on the different timeframes and socio-political contexts in which the two innovations of *they* emerged, it is likely that they show different patterns of use over time. While generic singular *they* likely acted as a pivot for the latter innovation, the two are not the same and will show different patterns of use and change.

=> I am proposing a typology that differentiates uses that are distinct in a discursive sense and likely to show different patterns of use over time. I am not going any further in differentiating specific uses because I don’t see that as a task for LLMs. That is ultimately at the heart of any investigation into changing uses of *they*. The role of LLMs is to filter first and thus enable humans to study *they* better.

2.3 My approach

Follows QA template described in Gan et al. 2024 and 2025. Reasoning for that is given in Gan et al. 2025, section 7.2.

2.4 LLMs as (linguistic) annotators

3 Data

First, we need to import all raw data by calling the relevant script. The raw data are shaped like so:

- Human participants have a dictionary each in which every data point, their annotation, and (optional) their comment is found.
 - Human participants are put into 5 blocks, blocks A-D and the group of pilot participants
 - LLMs come in two groups: commercial and open source
 - in each LLM group, there are different combinations of LLMs and prompting strategies, of which several runs may exist
 - each combination of LLM and prompting strategy is treated as a participant.
 - For every run of that combination, a dictionary is found in which every data point is found with their annotation and their full response saved like human annotators' comments
- Point about MA-era-dataset being limited (number of singular *they* examples; context-independence)

174 authentic uses of *they* (Reddit) Context-(in)dependent 56–59 for plural/generic/singular (do this dynamically & as a table)

While similar datasets for testing disambiguation skills such as Yuan et al. (2023) are only sentence-level, the present dataset has variable item length (from a few words to several paragraphs #TODO: get the min and max from the dataset) and the source of the item is always provided, allowing for deeper research, if necessary, to perform the task.

Each annotation instance was treated as an item-level trial for scoring and condition-wise breakdowns, while inferential comparisons were conducted on aggregated participant or run-level performance

Mention 10 intro examples that were cherry-picked for balance and had a precise reasoning behind the solution and were identical for all human participants. Clarify if those 10 examples were part of the test set for LLMs (shouldn't have been but need to ensure).

Randomly sampled for each participant to avoid bias

System of blocks with overlap

Point out that dataset is available

To illustrate, the variable name `commercial_llm_responses` has the entry “context-agnostic_zero-shot_claude-opus-4-1-20250805”, which tells us the prompting strategy (context-agnostic-zero-shot) and the LLM with its dated version (claude-opus-4-1-20250805). This entry has three runs saved. Each run is of the type “dict” and has 174 data points saved with their answers as keys and values. This number corresponds to the number of data points in the original test set.

4 TODO: figure that shows the annotation process as a flowchart in order to explain what tasks were being tested; point out that this applies to other datasets as well; explain that LLMs only mimic reasoning, but, depending on the task, fairly successfully

4.1 Participants

Local pilots were employed to avoid ethical concerns associated with crowdsourced labor and to maintain a controlled environment for evaluating task design and prompting strategies. However, nearly all available local participants were L2 speakers of English, an uncommon demographic for such experiments, especially seeing as they are chosen as experts, a status usually associated with native speakers. Therefore, two additional groups of participants were recruited through Prolific, one of which was composed of self-identified linguists who were also native speakers of English, and the other of non-linguists with English as L1. As such, the groups of human participants are either linguists or not, and either L1 or L2 speakers of English, or both. The results will thus be able to show, in addition to the human-LLM comparison, how the performances of different groups of human participants compare to each other.

All human participants were stratified into four blocks to manage study duration and mitigate participant fatigue effects while still maintaining a fairly high dataset size to ensure statistical power and diversity in the dataset. Each block received an overlapping set of 48 examples common to all participants, supplemented with block-specific items. Example distribution across blocks was randomized whilst maintaining within-block balance of antecedent types.

Blocks A–D comprised 11, 12, 10, and 8 participants respectively, with the pilot group comprising 7 participants. All participants annotated 48 examples, yielding a study duration of approximately one hour per participant.

4.1.1 Demographics

Women, age range (+ bibliographic justification)

4.1.2 Expertise

Explain issue with Prolific setup. Explain possibility of post-hoc culling the participants by attentiveness to still get appropriate data.

4.2 LLMs

4.2.1 Models

(so far) only the largest commercially available LLMs from the Claude & GPT family

Open Source:

Used this as my main guiding resource to tick the most important boxes: <https://www.bentoml.com/blog/navigating-the-world-of-open-source-large-language-models>

So far run: - Deepseek latest few models - Qwen3 latest few models - Momo V2 Flash (only non-free available on OpenRouter)

Fully local models make somewhat less sense for this kind of “piloting for the field” experiments as researchers having access to machines with 20+GB of RAM is less realistic than researchers getting access to computing clusters that can run far larger models. The latter is a more typical setup that you will see at research institutions and is often more readily financed than a machine that is only used by one person and costs double or triple of what their allocated funds typically allow.

Hence, the “local” category will comprise open source models run in a university-run computing cluster.

4.2.2 Prompting Strategies

Context-agnostic or with URL which only works for commercial models.

Invented examples were used as dev set to develop a prompting strategy.

The task description has brief examples interspersed, not fully-fledged ones.

Structured output reduced to just the last word. Worked perfectly.

5 Methods

Explain basic GT comparison

Justify choice of accuracy (F1 doesn't necessarily make sense over accuracy as the three classes are balanced. Also mention that MCC was considered.)

The annotation scheme was tested by a colleague with expertise in linguistics but no direct experience in my specific research area, and the IAA was found to be good (Cohen's kappa of 0.83 which is considered almost perfect agreement).

Confirmatory and quasi-experimental study

6 Results

6.1 Hypothesis 1

The first hypothesis aims to test if any LLM-based approach would yield the highest overall performance.

Table 1: The ten highest-performing participants by accuracy

Participant Group	Accuracy
block_Pilot	96%
context-via-permalink_google-gemini-3-pro-preview	92%
context-via-permalink_gpt-5	91%
context-ondemand_zero-shot_gpt-5	90%
block_A	90%
block_B	90%
block_B	90%
block_C	90%
block_D	90%
block_D	90%
context-agnostic_zero-shot_gpt-5	89%
context-via-permalink_gpt-5-2025-08-07_with-justification-unspecified	88%
block_C	88%
block_Pilot	88%
Non-Linguist Human Nr. 18	88%
context-via-permalink_deepseek-deepseek-v3.2-speciale	87%
context-via-permalink_claude-opus-4-1-20250805	86%
context-agnostic_zero-shot_z-ai-glm-5.1	86%
majority_vote	86%

Table 1: The ten highest-performing participants by accuracy

Participant Group	Accuracy
context-via-permalink_gpt-5-2025-08-07_with-justification-conservative	86%
context-agnostic_zero-shot_claude-opus-4-1-20250805	85%
block_B	85%
context-via-permalink_claude-sonnet-4-20250514	84%
context-via-permalink_z-ai-glm-4.7	84%
block_B	83%

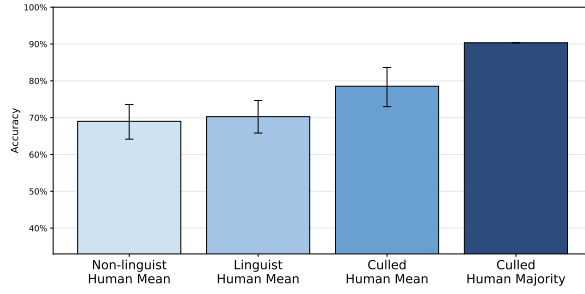
6.2 Hypothesis 2

Table 2: Overall accuracy by three human groups

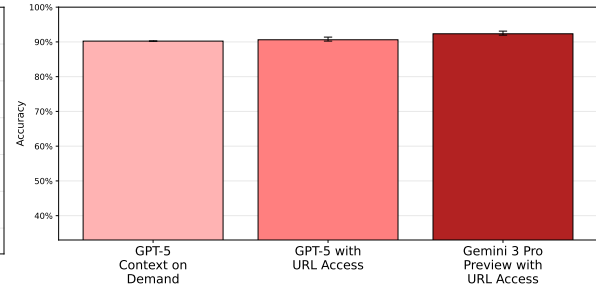
Group	Accuracy
Non-Pilot Humans	68.5%
Pilot Humans	80.8%
Non-Linguists	69.0%

The difference between the crowdworkers and the local student pilot participants is not statistically significant ($p = 0.059$, $\alpha = 0.05$). For inferential testing in this section, only Non-Pilot Humans and Pilot Humans were compared. The Non-Linguists group is reported descriptively for context.

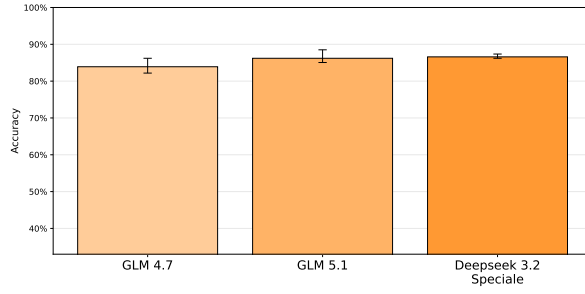
The group of culled participants has 20 participants, of which two are local students and 18 are crowdworkers.



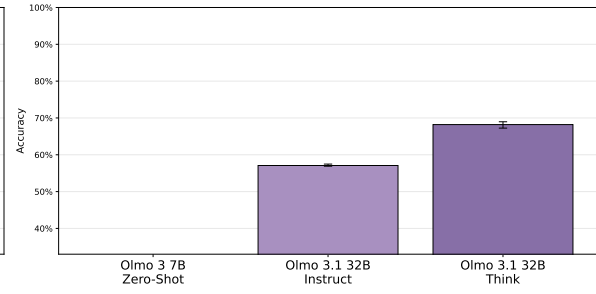
(a) Human Group Accuracies



(b) Commercial LLM Accuracies



(c) Open Source LLM Accuracies



(d) Local LLM Accuracies

Figure 1: Participant accuracies with 95% confidence interval

6.3 Results by label

6.3.1 Singular vs Plural

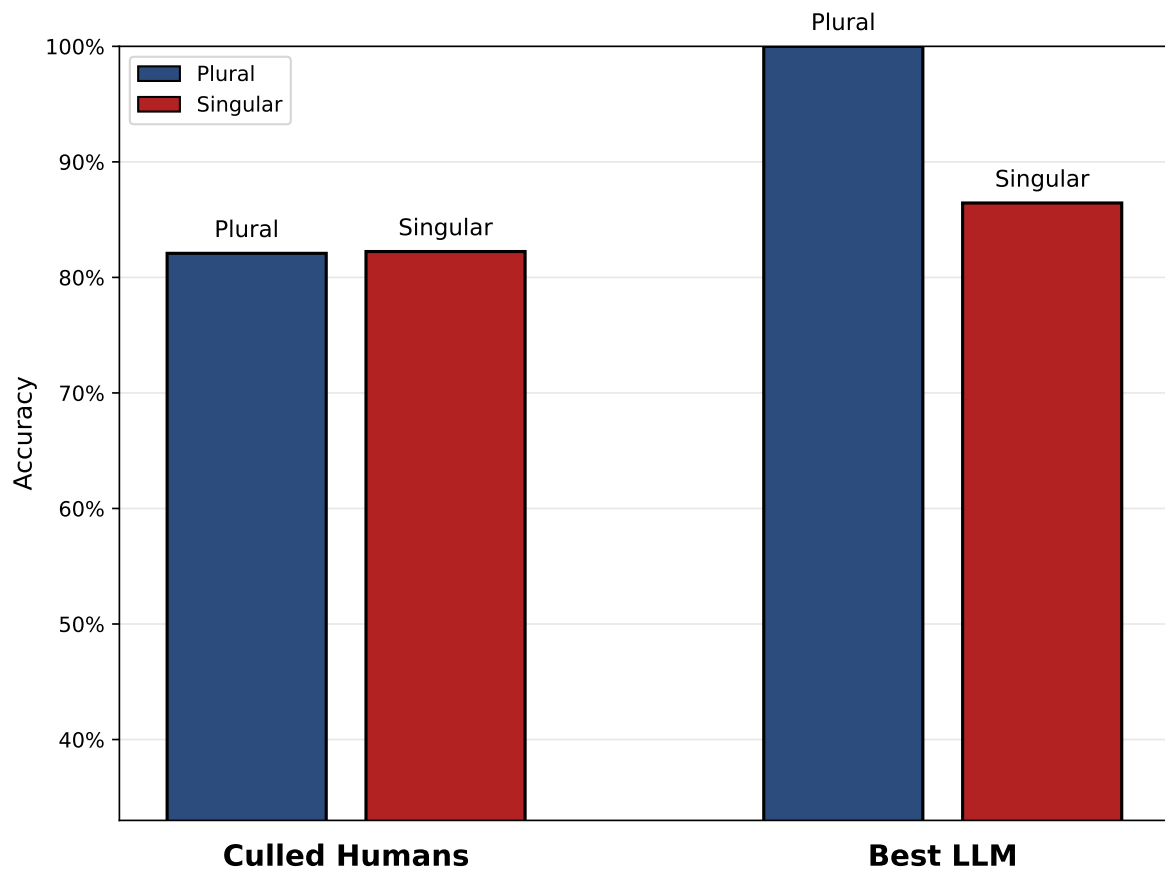


Figure 2: Accuracy difference between plural/singular *they* and LLMs/humans

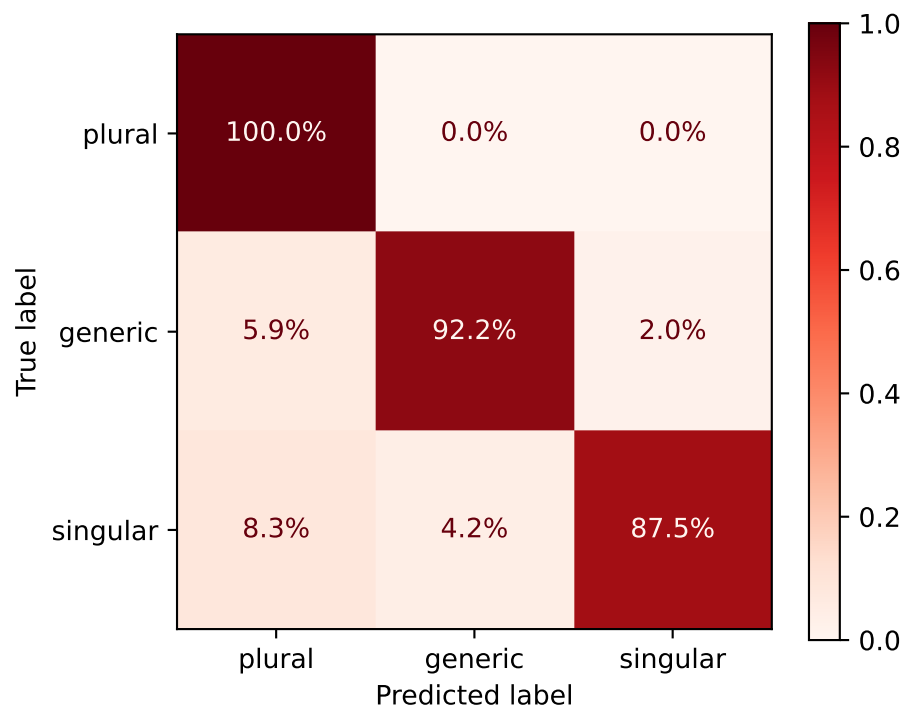


Figure 3: Confusion matrix for the best LLM

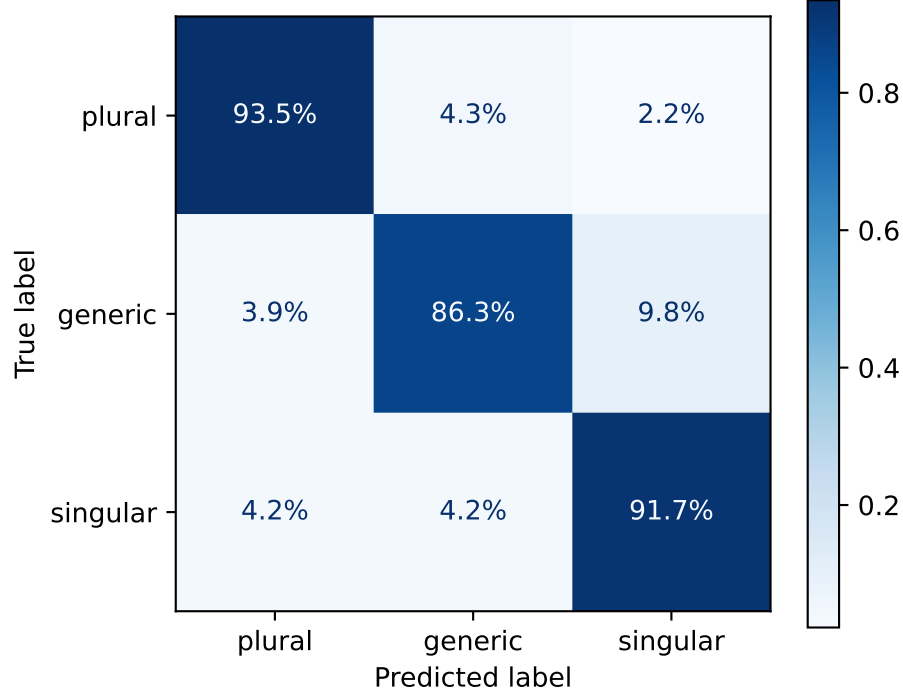


Figure 4: Confusion matrix for culled human mean

TOST (independent samples, Welch)

- $\alpha = 0.050$, $\Delta = 0.020$
- $t_{\text{lower}}=2.232$, $p_{\text{lower}}=0.0189$ (NI decision uses this)
- $t_{\text{upper}}=0.792$, $p_{\text{upper}}=0.7810$

Decision: Non-inferiority **established**.

6.3.2 Context-(in)dependence

Table 3: Accuracy for context-dependent and context-independent items for two different LLM strategies and the culled human mean

Group	Context-Dependent	Context-Independent
Culled Human Mean	90.0%	77.5%
GPT-5 Context-Agnostic	79.4%	92.9%
GPT-5 with URL Access	83.3%	94.5%

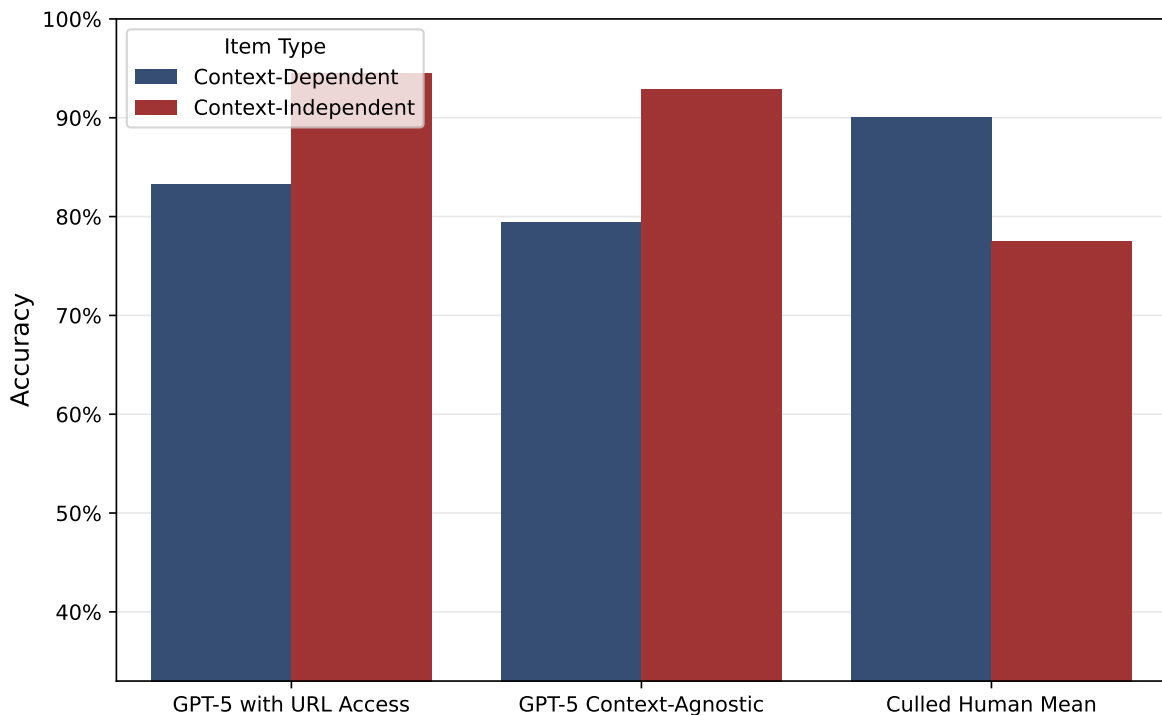


Figure 5: Accuracy for context-dependent and context-independent items for two different LLM strategies and the culled human mean

When looking at the behavior of GPT-5 with URL access, it shows a difference between my own expectations of what examples required context for disambiguation and which ones did not. Of the 34 items that I marked as context-dependent, less than half (14) caused the model to access the context via provided URL. In turn, of the 25 items that did cause the model to check context, only 14 were marked as context-dependent by me. Among the cases that the model failed to look up context I had marked as necessary, however, its accuracy at disambiguating the meaning of *they* still lay at a relatively high 80.0%, which is very close to the average accuracy of my local study pilots over all items, 80.8%. Among the cases that the model had correctly decided to look up context, its accuracy was only 76%, showing that it likely lacks capabilities in navigating heterogenous and complex contexts. This means that the model’s behavior does not fully align with my expectations of what items require context for disambiguation, and it also accesses context for some items that I would have considered context-independent. It is not clear what causes this difference in behavior but it does point to LLMs’ reasoning capabilities being different from human reasoning, generally to the detriment of the results. In light of study design, it suggests that LLMs fare better without having to research information on their own in order to complete a task, as noticing the lack of necessary information seems to be still too great a challenge for such models. This may impede the study of especially complex linguistic phenomena that require contextual and

semantic understanding of text.

When not specifying that the model should access context only in doubtful cases, thus leaving it up to the model, there is a clear tendency for the model to access context more often. Namely, of the 34 context-dependent cases 28 were checked for context, which is a recall of 82%. However, the model checked a total of 106 cases for their context, of which only 28 were actually context-dependent, meaning that the precision of the model's decision to check context was only 26%. This means that while the model is relatively good at recalling cases that require context, it also checks for context in many cases where it is not necessary, which may lead to worse performance due to the model's difficulties in navigating complex contexts. Indeed, in the specific context of this study, there was a measurable drop in accuracy by 10% (from 94% to 84%) when the model accessed context vs when it did not. For this reason, the same model performed better when it did not have the option to access context (89%) than when it did (86%), as shown in the previous section.

6.3.3 Performance with innovative forms of *they*

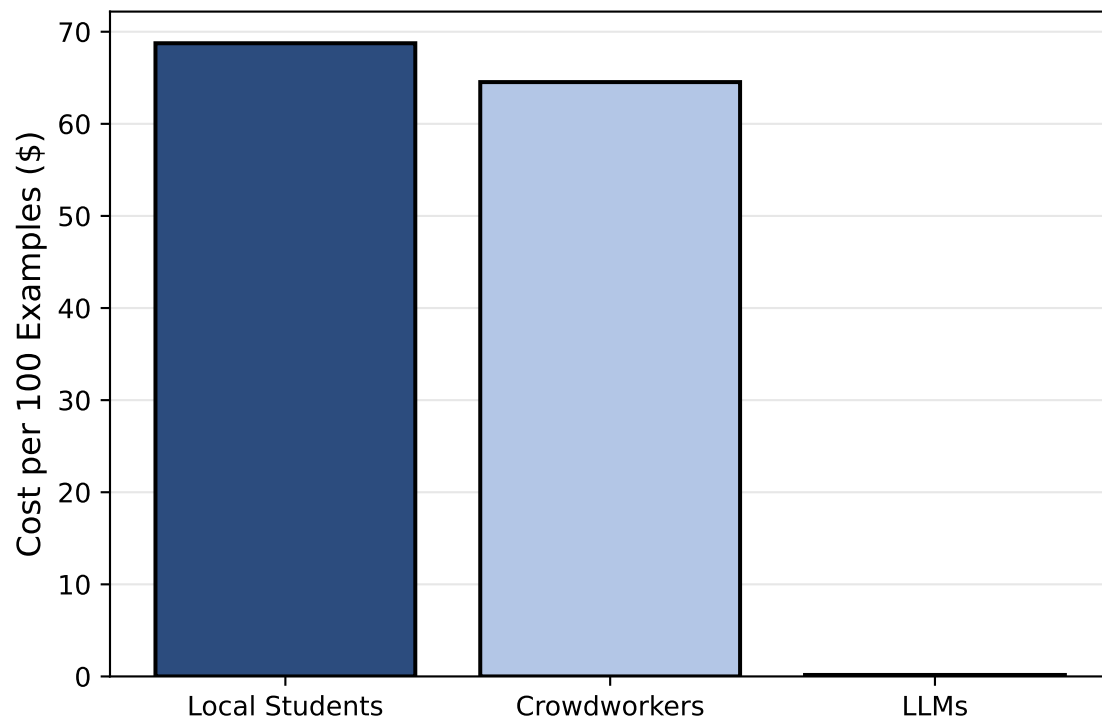
Since I care about recall of sg. and gen. *they* for the application in later research, the recall for those two should be featured apart for humans & LLMs

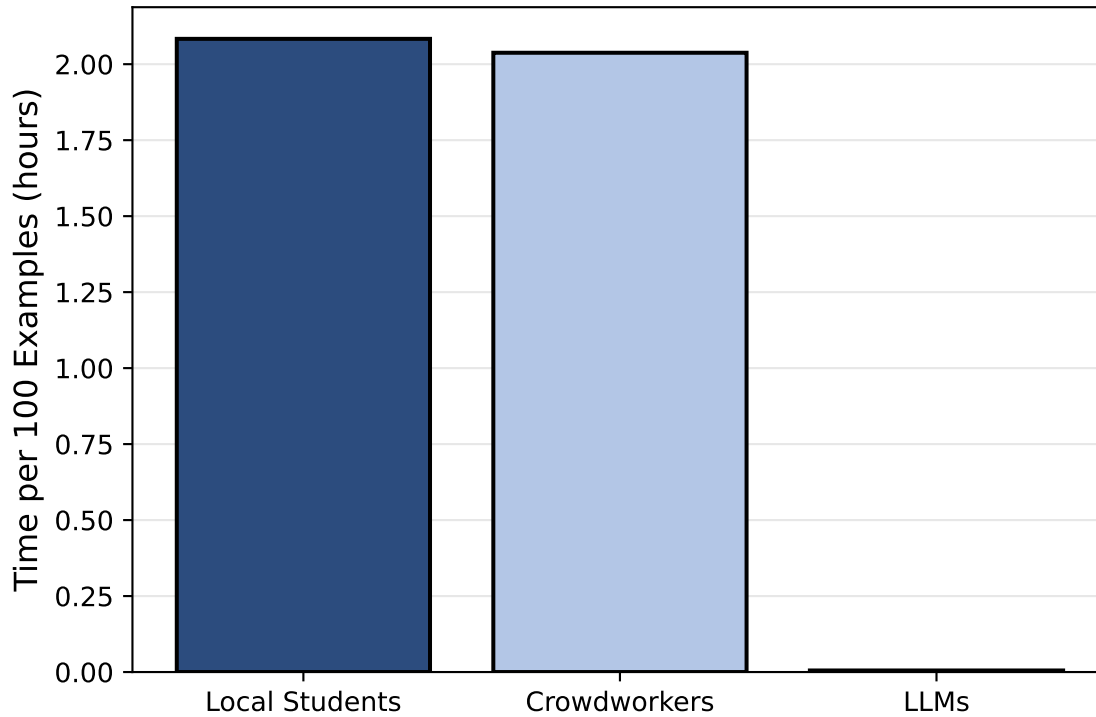
6.3.4 Projection onto naturalistic data

It might make sense to apply the per-class performance measures to a hypothetical test set with naturalistic frequency splits (take from own data) to then generate MCC and get an estimation of how well the different participants would do on naturalistic data. This will address the lingering question if the artificial selection of data points skewed the results or hid anything about the truth

This will be an important point for the conclusion.

6.3.5 Cost





- The cost of crowdworkers was inflated in this study due to filtering inattentive participants only post-hoc and not being able to tell the meaning of “educated in English language” before carrying out the study. Appropriate numbers were obtained thanks to the culling. This means in turn that the responses of 56.1% of crowdworkers were disregarded and their cost was spent fruitlessly.

7 Discussion

7.1 L1 Crowdworkers vs Local L2 Students

The results showed that this task in particular is not automatically solved better by native speakers than by informed L2 speakers. The latter performed at 81.0% as opposed to the 68.0% achieved by native speakers. This is not solely due to the L1 vs L2 status but also a question of competence. The local students had been selected based on their previous high academic achievements and specific knowledge of English linguistics of gender. The difference in performance is not, however, statistically significant. Still, it shows that at this level the native speaker status should no longer be taken as a prerequisite for (annotation) work in this and similar fields or activities. Local students are also preferable due to ethical and practical reasons as communication is easier and there are better mechanisms in place to ensure fairness and research ethics. (Alternative AI suggestion for more adequate phrasing: In this sample,

local L2 student pilots showed similar or higher mean accuracy than non-pilot crowdworkers. Because this comparison is observational and groups differ in selection and expertise, it should not be interpreted as an isolated L1 versus L2 effect. The result instead supports the practical point that trained local student annotators can be competitive with crowdworker annotators in this task. And: These data do not support treating native-speaker status alone as a sufficient criterion for annotator selection in this task.) ## Hypothesis 1

Table 1 reveals that the group of top performers is populated by more humans than LLMs, although most top performances cluster around the 90% mark. To directly address Hypothesis 1, the top performer is not an LLM but indeed a human. This participant was furthermore from the pilot participants recruited from the local student body. Hypothesis 1 is therefore rejected.

7.2 Hypothesis 2

The second Hypothesis aims to test if LLM accuracy at the task outperforms the mean human accuracy. As the overall goal of this research is to assess if LLMs could plausibly replace proper student assistants, it makes sense to define “mean human accuracy” more closely. The participants were recruited more liberally than a standard selection process for a student assistant position. They were recruited from only two seminar groups and were prioritized by availability. The crowd workers self-selected by ticking a box that said they had an education in “English language”, without specifying their level of expertise and, in some cases, not actually having studied linguistics, judging by their answers to the screening. To approximate what level of performance one might expect from an actual student assistant, I additionally report a culling-based benchmark variant as sensitivity analysis. This benchmark is defined by removing inattentive annotators and retaining the top half of the remainder. As this benchmark is interpretive and operational and not design-neutral, its results should be interpreted cautiously. still, it may serve as an estimate for what might be “true human performance.” Inattentiveness is formalized in terms of context-dependent performance, i.e. if their performance is at or below 33% when context is necessary to understand an example.

Figure 1 reveals that there is a marked difference between the overall and the culled human mean. This shows once again that crowdworkers as a whole are not a reliable resource for annotation work. The culled human mean, which is supposed to approach the performance level of a specialized student assistant, still falls short of the best three LLMs’ performance. There, it seems that the prompting strategy had only a minor impact, improving performance by just a few percent. The strongest combination of LLM and prompting strategy, {‘label’: ‘Gemini 3 Pro Preview with URL Access’, ‘value’: 0.9233716475095785}, shows a performance that is , however, not significantly better than the culled human mean at $p = 0.480682373046875$, $\alpha = 0.05$ according to a McNemar’s test. Hypothesis 2 can therefore not be confirmed. It is still remarkable that the best result of the study is achieved by taking the majority vote of the culled human group. Applying this to a real-world setting, it suggests that teams of experts are still the strongest at this kind of annotation task.

7.3 Hypothesis 3

Hypothesis 3 aims to show non-inferiority of LLMs at

7.4 Research question 2

In an attempt to assess the problem-solving behaviors qualitatively, three aspects of human and LLM responses were analyzed: the optional comment fields in the human survey, the LLMs’ reasoning steps, and the accuracy of LLMs in determining context-dependency in the items.

Human-supplied comments were analyzed to reveal the strengths and weaknesses of their reasoning. The following themes were identified: the biggest issue was their understanding of the in-between category of generic *they*, followed by some minor cognitive barriers. Their biggest strength seemed to be their skill in discerning when they needed additional context to make a decision (though the quantitative results show that this did not give them an edge over LLMs). The cognitive barrier for humans in this task lies in the processing of very long input texts. While most items in the dataset are fewer than ten sentences long, item 23 is 38 sentences long and the pronoun in question is located in the penultimate sentence. On top of that, the text uses several uncommon words and therefore requires additional research for most. Human participants, potentially also fatigued when they encounter this item, have to choose how much context they read. In all likelihood, they do not want to read such a lengthy text. LLMs, on the other hand, are much less sensitive to the length of text, as long as the input is well within their window (#SOURCE). Indeed, humans answered wrong N times while LLMs only M (#TODO). As previously discussed (#REF), the category of generic is neither plural nor singular. It is not taught as standard grammar teaching to laypeople and not even linguists are assured to have come across it in their studies or work. Thus, some items in the dataset caused trouble for the human participants. For instance, item 20 shows *they* referring to hypothetical football players, a plural referent. However, the hypothetical nature of the statement seemed more salient to the participants than the plural number of the referent, leading them to classify it as generic. Similarly, item 134 shows *they* referring to *your ancestor*, which is an impersonal use of the second person possessive pronoun *yo*, thus making the reference generic. Most human reasonings focus on the use of *your* but fail to understand its impersonal meaning. The biggest issue with the generic category for human participants were the cases of *they* referring to *everyone* and *noone*. According to the annotation instructions they were to annotate these as plural. The instructions explicitly referred to *everyone*, *noone*, and *nobody* for the plural category. Instead, human participants seemed to have retained the instruction that statements “that [apply] to all X” and accordingly classified these items as generic. This reveals a crucial difference between human and LLM annotators. The latter famously exceed at repeating patterns and are thus strong at recognizing cases they received instructions for, regardless of any potential weaknesses in the internal logic of the instructions. Humans, on the other hand, seem to grasp the underlying logic of instructions and override

inconsistent instructions. Had the participants been in a real-world annotation setting, they would have likely asked for clarification on the instructions, thus revealing this inconsistency and helping to improve my annotation scheme. Their quantitative weakness (i.e. incorrectly annotating examples involving *everyone* and *noone* according to the instructions) is thus really a strength, i.e. because they understood the faults of the underlying logic. Though correct usage of an LLM may have also brought this weakness to the forefront.

When analyzing the LLMs' reasoning steps, #TODO

- problems unique to LLMs:

** maybe umkao example and how LLMs made more base assumptions in that case

- strengths unique to LLMs:

Some errors were shared among human and LLM participants, #TODO

To determine the LLMs' accuracy in identifying context-dependency, the prompting strategy that includes the context URL was modified to additionally ask the LLM to declare whenever it decided to access the

7.5 Limitations

Because inferential analyses were primarily based on aggregated accuracies rather than a full item-level hierarchical model, residual item-difficulty effects may remain.

Fully ambiguous cases (even when researching context etc) were not included in this study.

Other languages have their own examples of gender-neutral language with different properties and etymologies/histories. This study may or may not be generalizable to those cases.

I did not investigate a human-LLM cooperative approach and how it would compare to either group alone. This would be an interesting avenue for future research, especially given the complementary strengths but also weaknesses of humans and LLMs revealed in the qualitative analysis. And not least because public discourse often proposes the idea that humans are not threatened by LLMs as long as they work together, a claim worth testing in practice.

8 Conclusion

The results are mixed and it is not straightforward to say if LLMs have truly attained human level at this particular task.

- Results show that semantically complex annotation in linguistics may be done by LLMs at much cheaper cost. The human-machine margin in cost arguably more than compensates the margin in quality.

- The results also show that the under-the-hood reasoning of LLMs has already diminished the importance of prompt engineering for this task, further reducing the risk of human variability and reducing the importance of human expertise altogether.
- Potentially applicable to any other variable dependent on situational, pragmatic, ... context, i.e. ‘distant reading’
- Similar research tends to confirm this finding.
- However, some cursory analysis of LLM errors shows that LLMs are prone to being fooled by context.
- Also, bias towards standard English still a limitation, i.e. innovative forms like specific singular *they* at disadvantage.
 - Reasoning models are likely to become better for a while before hitting a plateau.

Ich nehme an, dass LLMs weite Teile der aktuellen linguistischen Forschung tangieren oder massgeblich prägen werden. Dabei befürchte ich aber, dass es zu Problemen verschiedener Art kommen wird. In etwa, weil die Fähigkeiten der LLMs überschätzt oder falsch eingeordnet werden. Oder weil sich, in Anlehnung an den Papst, zu viel Macht und/oder Wissen in den Händen weniger Firmen sammeln wird. Letzteres führt zu einem Plädoyer für wissenschaftsorientierte Open-Weight-Modelle.

- To do:
 - Further qualitative analysis of human & LLM responses
 - Application and evaluation of open source LLMs (replicability!), N-shot prompting, fine-tuning
 - Analyze uses of they in various corpora, potentially uncovering the history of its change and diffusion – Threat to entry-level / student assistant positions?
 - Environmental impact concerns?

8.1 Debugging etc.

References

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